

Homework 12

Dec 3, 2025

1 (30'). Let

$$\text{INDSET} = \{(G, k) \mid G \text{ has an independent set of size } k\},$$

and

$$\text{CLIQUE} = \{(G, k) \mid G \text{ has a } K_k \text{ subgraph}\},$$

where the undirected graph G is represented by its adjacency matrix.

Prove that $\text{INDSET} \leq_p \text{CLIQUE}$.

2 (30'). A *2-CNF* formula is a conjunctive normal form in which each clause contains exactly 2 literals. For example,

$$(x \vee y) \wedge (x \vee \neg z) \wedge (\neg y \vee z)$$

is a 2-CNF formula.

Let 2-SAT denote the language consisting of (encodings of) all satisfiable 2-CNF formulas. Show that 2-SAT is in P.

3 (40'). Let

$$L_{\text{halt}} = \{(\alpha, x) : \text{Turing machine } M_\alpha \text{ halts on input } x\}.$$

Show that L_{halt} is NP-hard.